Designing and Building File-Folder Bridges as an Introduction to Engineering

The Waddell A-Truss Bridge

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Objectives

- Learn about structural engineering:
  - Through a hands-on bridge-building project.
  - Through the use of free computer software.
- Learn about the ongoing West Point Bridge Design Contest.

A Typical Bridge-Building Project

- Students receive a pile of Popsicle sticks and some glue.
- Students build a bridge, based on...
  - A picture.
  - A vague idea of what a bridge should look like.
- Bridges are weighed.
- Bridges are tested to failure.
- Highest strength-to-weight ratio wins.

What do students actually learn from this experience?

What They Don't Learn

The Essential Characteristics Of Engineering

- A systematic design process precedes construction.
- Engineers design; Contractors build.
- The design process is informed by math and science.
- Design is iterative.
- Structures are designed to carry code-specified loads safely and economically.
  - Designed to stand up, not to fail.
  - Strength-to-weight ratio is *never* the objective.

Why File Folders?

- Inexpensive.
- Easy to cut, bend, and glue.
- Surprisingly predictable structural behavior.
- Can be used to build:
  - Tubes and bars.
  - Connections that are stronger than the attached structural members.

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Our Agenda

- Introduction to Truss Bridges
- *Start building a truss*
- Forces and equilibrium
- *Continue building the truss*
- Structural analysis
- *Finish the truss*
- Materials testing
- Structural evaluation
- Structural design
  - Manual method
  - Using the West Point Bridge Designer

This allows time for the glue to dry

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What You Need to Know

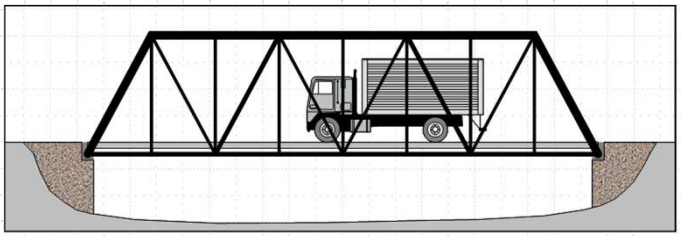
- For building a file-folder bridge:
  - NONE
- For analyzing a file-folder bridge:
  - Basic algebra
  - Geometry – Pythagorean Theorem
  - Trigonometry – sine and cosine
  - Physics – forces, equilibrium
  - Computers – spreadsheets
- For the West Point Bridge Designer
  - NONE

These concepts could be taught in the context of this project

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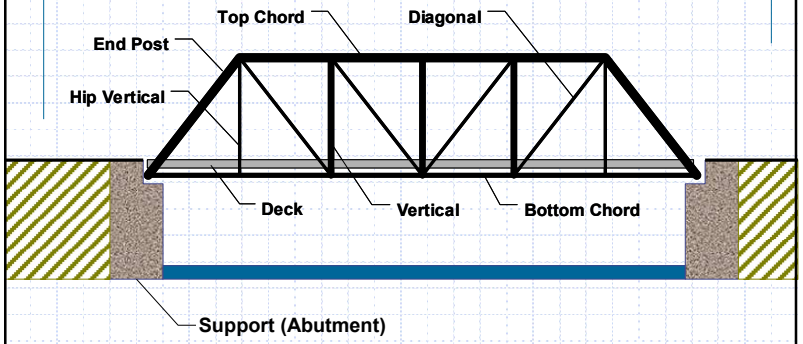
What is a Truss?

- A structure composed of members connected together to form a rigid framework.
- Usually composed of interconnected triangles.
- Members carry load in tension or compression.



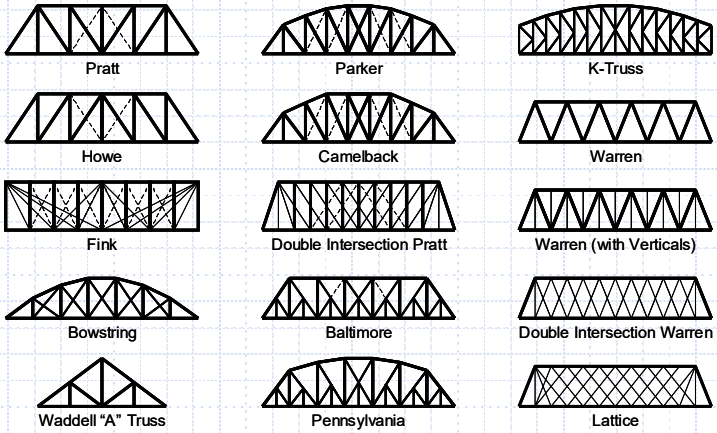
8

Component Parts



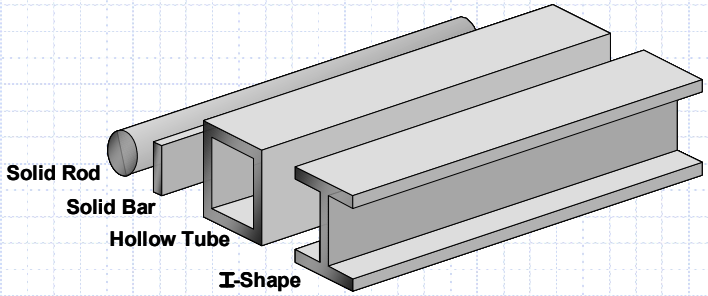
9

Standard Truss Configurations



10

Types of Structural Members

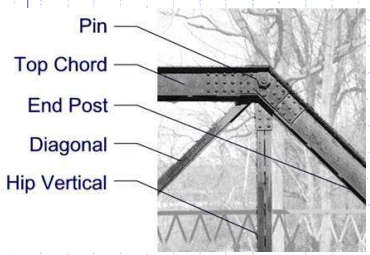


These shapes are called cross-sections.

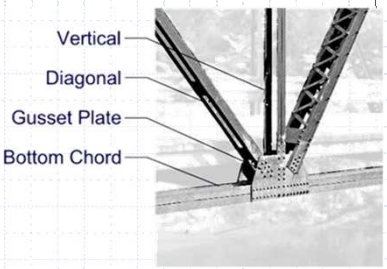
11

Types of Truss Connections

Pinned Connection



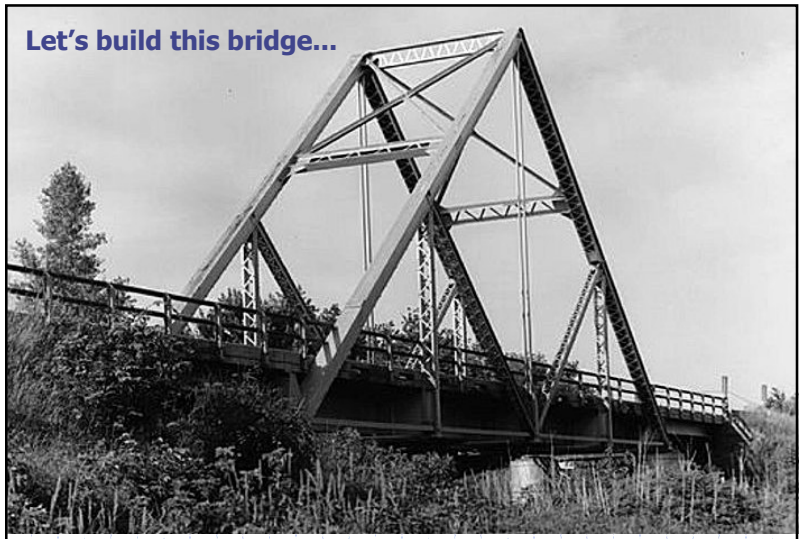
Gusset Plate Connection



Most modern bridges use gusset plate connections

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Let's build this bridge...

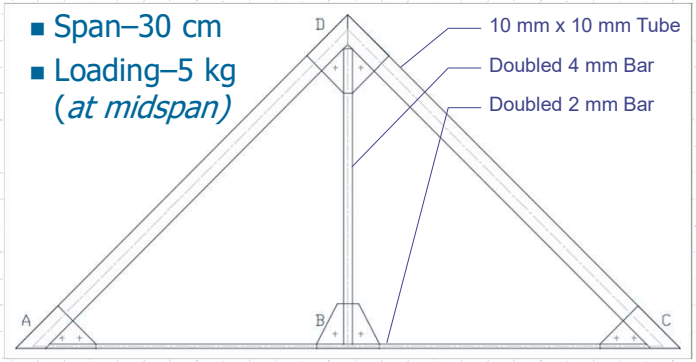


Waddel "A Truss" Bridge over Lin Branch Creek
Trimble, MO

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The Design

- Design Requirements:
  - Span–30 cm
  - Loading–5 kg (*at midspan*)

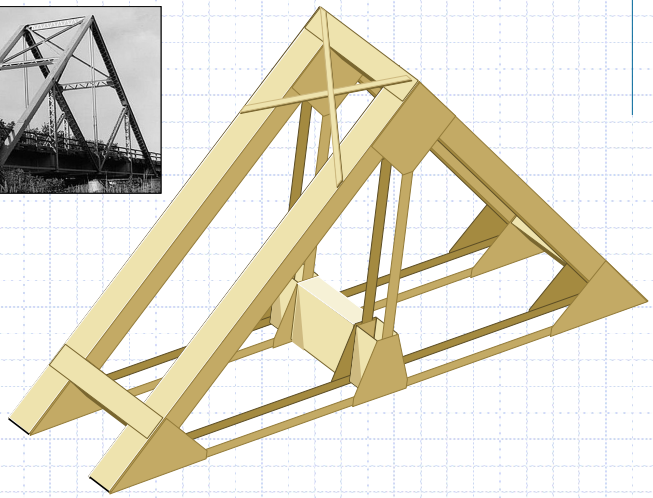


We'll talk about how it was designed later...

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Our A-Truss Bridge





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Materials & Equipment

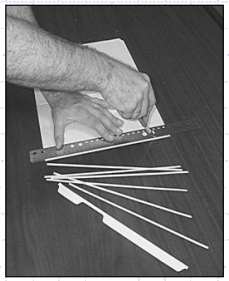
- File folders
- Yellow carpenter's glue
- Building board (Styrofoam or cork)
- Pins
- Scissors
- Metal ruler*
- Hobby knife or single-edge razor blade*
- Rubber cement*

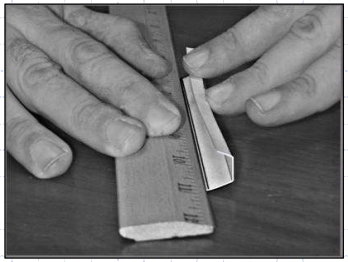
*Required only for prefabrication of structural members

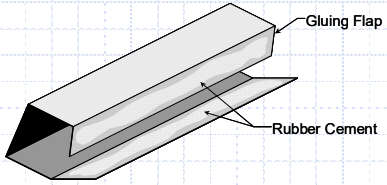
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Prefabrication of Members

- Cut out bars
- Cut out and assemble tubes
- Cut out gusset plates
- Trim bars and tubes to length



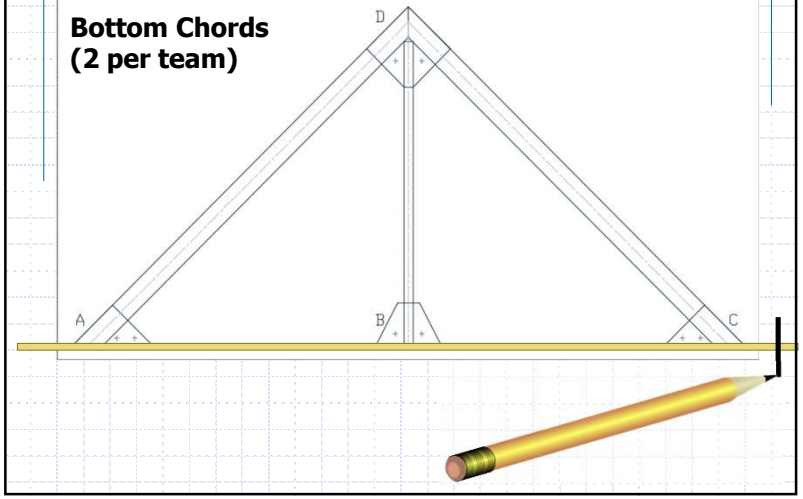




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Trim Bars and Tubes to Length

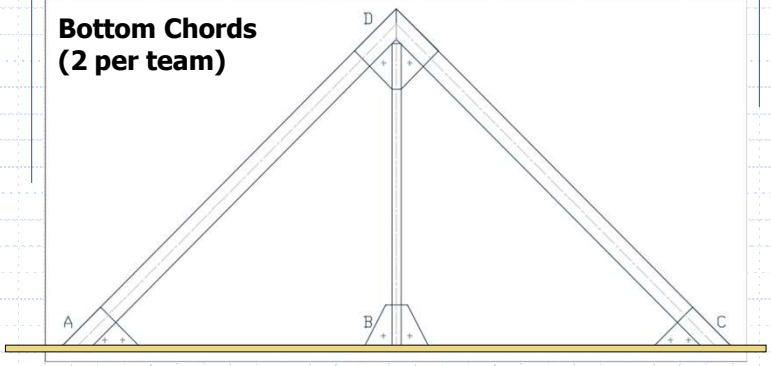
Bottom Chords
(2 per team)



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Trim Bars and Tubes to Length

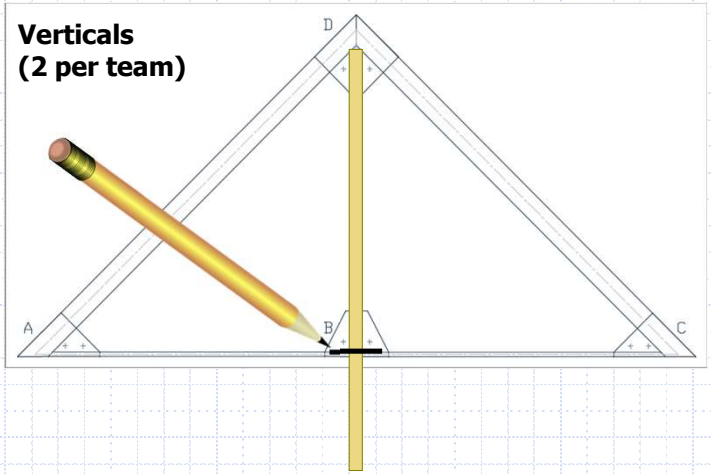
Bottom Chords
(2 per team)



19

Trim Bars and Tubes to Length

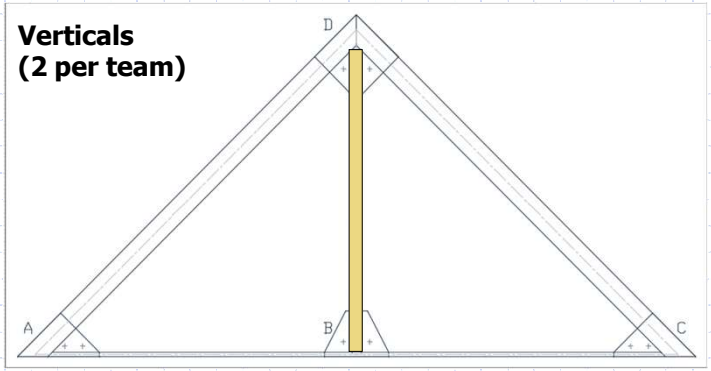
Verticals
(2 per team)



20

Trim Bars and Tubes to Length

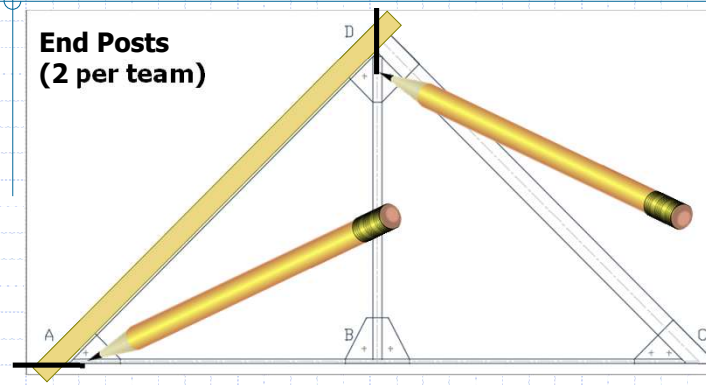
Verticals
(2 per team)



21

Trim Bars and Tubes to Length

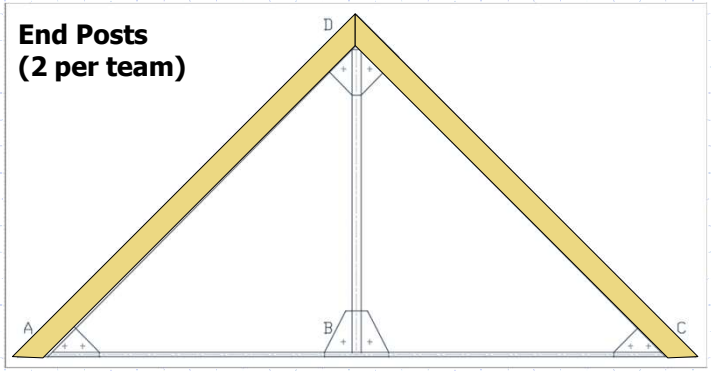
End Posts
(2 per team)



22

Trim Bars and Tubes to Length

End Posts
(2 per team)

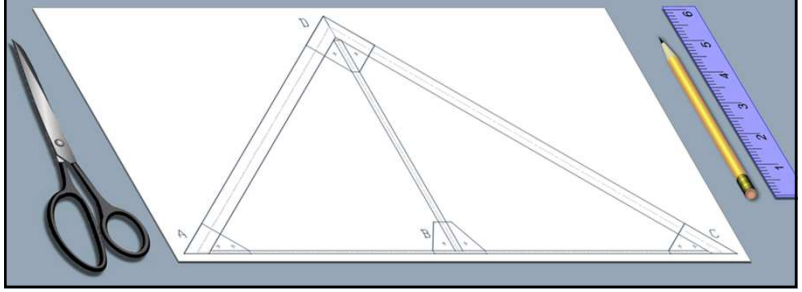


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Set up the Building Board

Each Team Member:

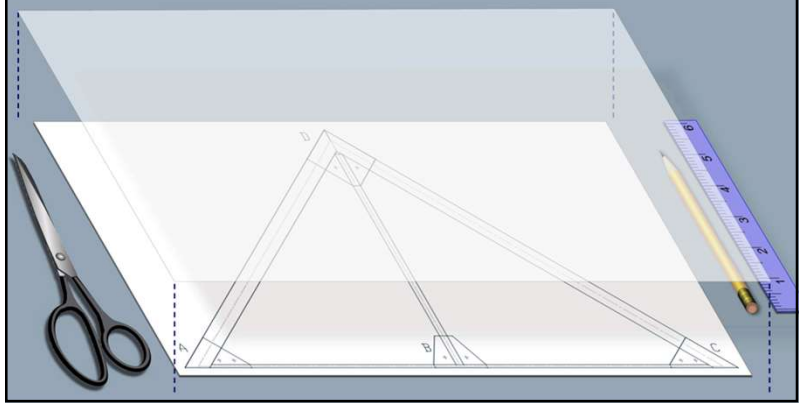
- Place the layout drawing on your building board.



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Set up the Building Board

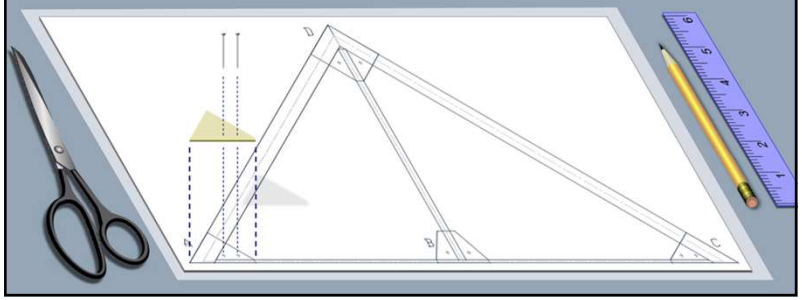
- Place a sheet of plastic wrap over the layout drawing.



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Add Gusset Plates

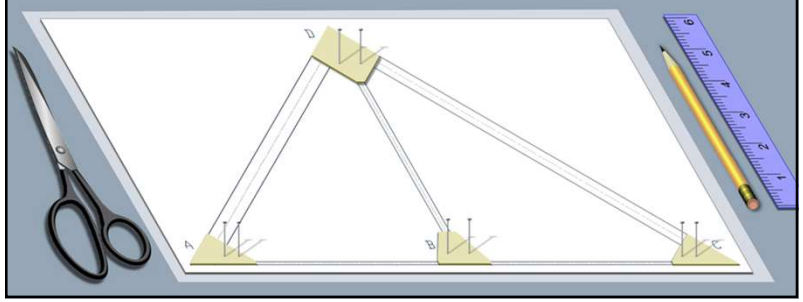
- Place Gusset Plate A at its correct location on the layout drawings.
- Hold it in place with two pins.



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Add Gusset Plates

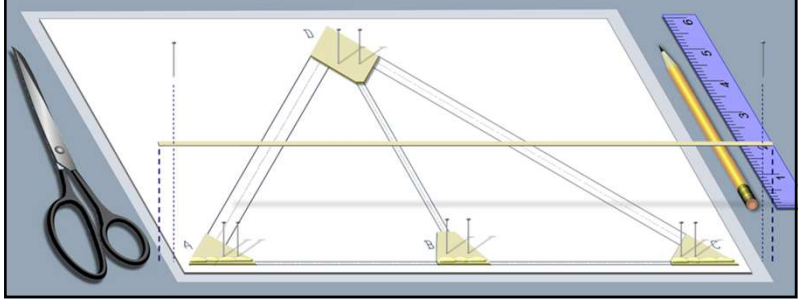
- Repeat the process for Gusset Plates B, C, and D.



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Add Bars

- Apply a line of glue along the bottom edge of Gusset Plates A, B, and C.
- Place a 2 mm bar in position as the bottom chord AC.
- Stretch tight and hold in place with two pins.

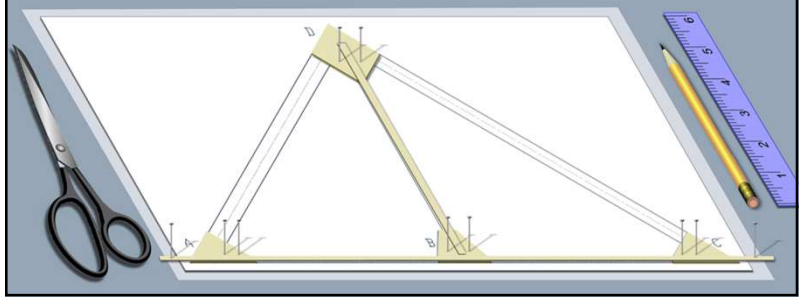


28

Add Bars

- Apply glue to Gusset Plates B and D.
- Place a 4 mm bar in position as the vertical member BD.
- Stretch tight and hold in place with your fingers.

Each team should now have *two* of these subassemblies — the *lower half* and the *upper half* of one truss.

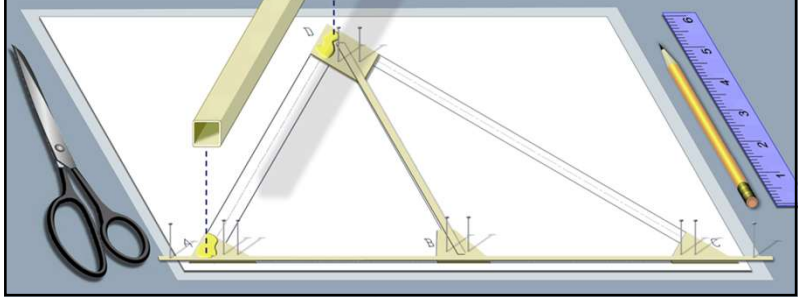


29

Add Tubes

For the *bottom half* of the truss (one per team):

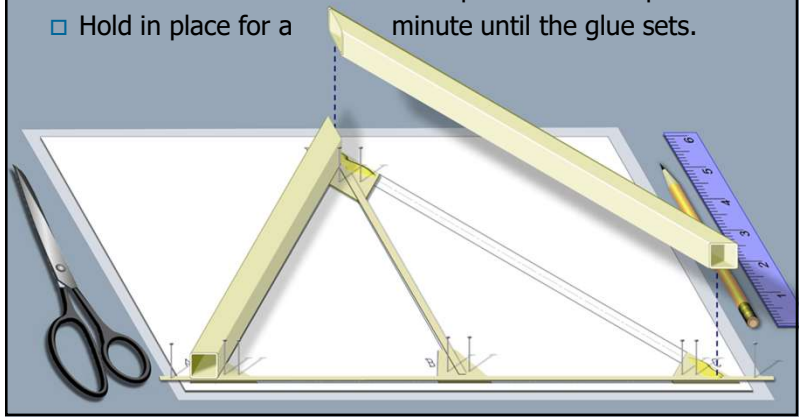
- Apply glue to Gusset Plates A and D.
- Place a 10mm x 10mm tube in position as end post AD.
- Hold in place for a minute until the glue sets.



30

Add Tubes

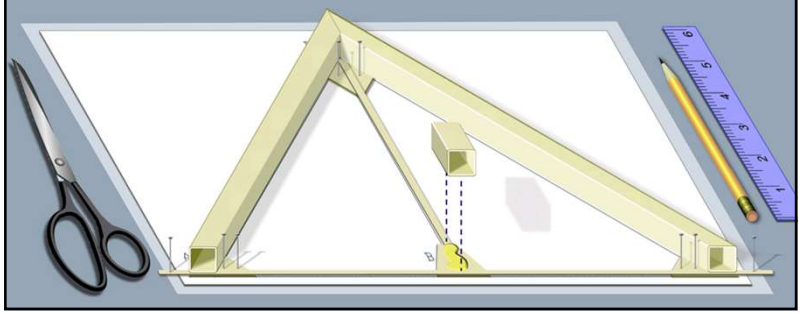
- Apply glue to Gusset Plates C and D.
- Place a 10 mm x 10 mm tube in position as end post AD.
- Hold in place for a minute until the glue sets.



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Add Tubes

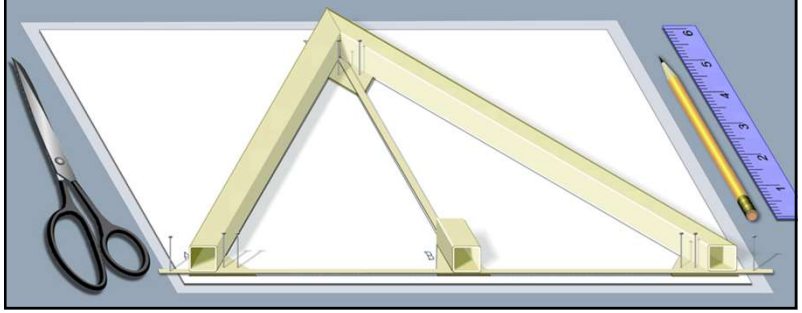
- Cut a 2 cm length of 10 mm x 10 mm tube.
- Apply glue to Gusset Plate B.
- Place the tube vertically on the gusset plate.
- Hold in place for a minute until the glue sets.



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The Finished Half-Truss

- Allow all glue joints to dry.



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Forces, Loads, & Reactions

- Force – A push or pull.
- Load – A force applied to a structure.

Self-weight of structure, weight of vehicles, pedestrians, snow, wind, etc.

- Reaction – A force developed at the support of a structure to keep that structure in equilibrium.

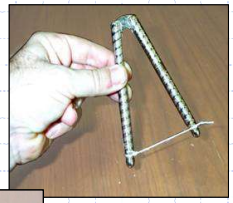
Forces are represented mathematically as VECTORS.

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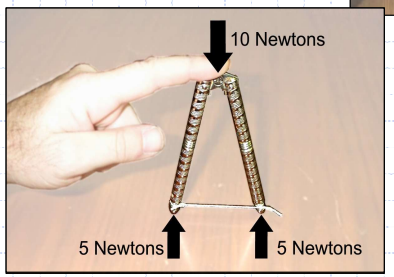
Equilibrium

Newton's First Law:

An object at rest will remain at rest, provided it is not acted upon by an unbalanced force.



A Load...



...and Reactions

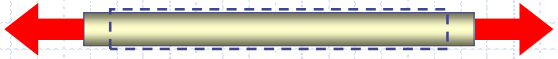
35

Tension and Compression

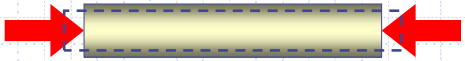
An unloaded member experiences no deformation



Tension causes a member to get longer

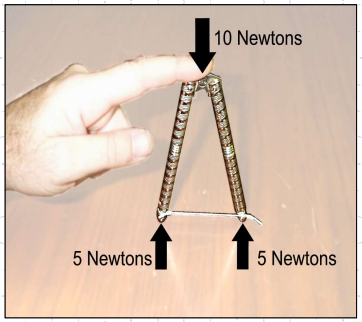


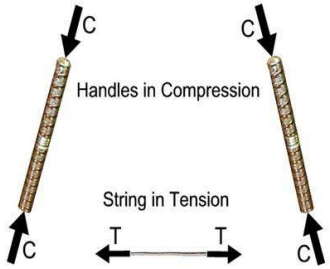
Compression causes a member to shorten



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Tension and Compression



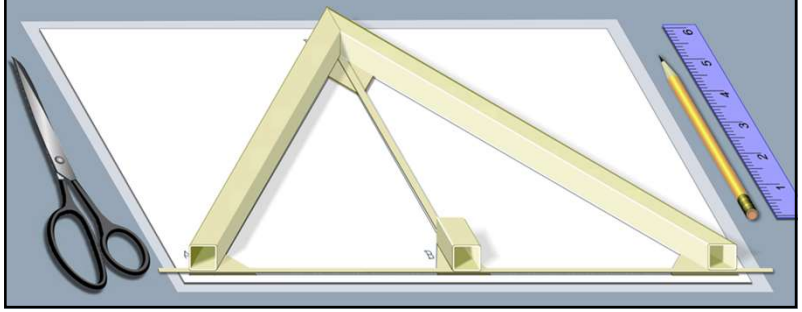


EXTERNAL FORCES and INTERNAL FORCES
Must be in equilibrium with each other.

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Assemble the Two Halves

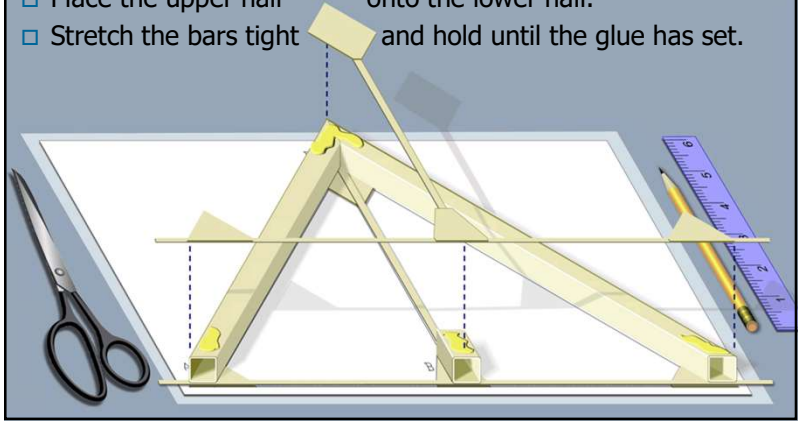
- Pull out all of the pins on both halves of the truss.
- *Carefully* separate the upper half of the truss from the plastic wrap.
- Keep the lower half of the truss on the building board.



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Assemble the Two Halves

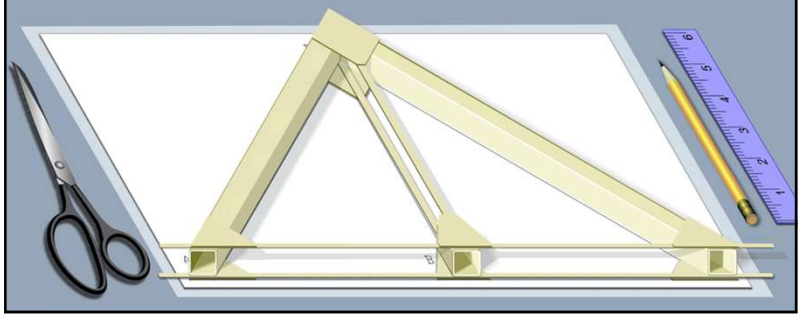
- Put glue on the tubes at A, B, C, and D.
- Place the upper half onto the lower half.
- Stretch the bars tight and hold until the glue has set.



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Assemble the Two Halves

- Allow all glue joints on the completed truss to dry.



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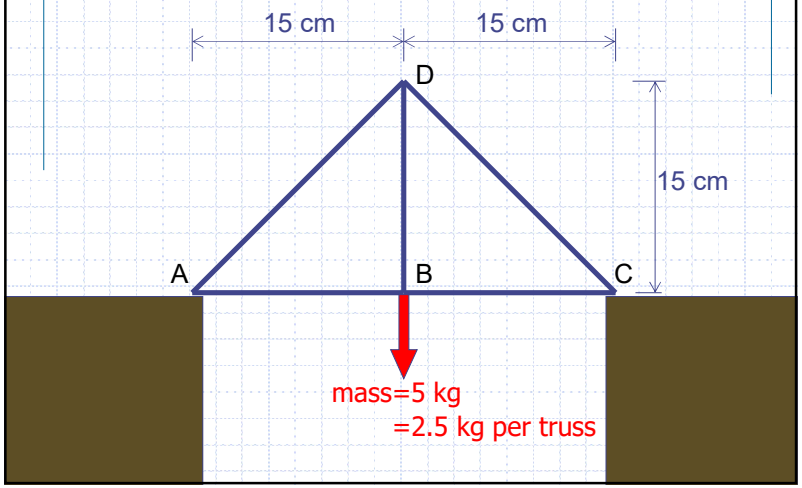
Structural Analysis

- For a given load, find the internal forces (tension and compression) in all members.
- Why?
- Procedure:
  - Model the structure:
    - Define supports
    - Define loads
    - Draw a free body diagram.
  - Calculate reactions.
  - Calculate internal forces using "Method of Joints."



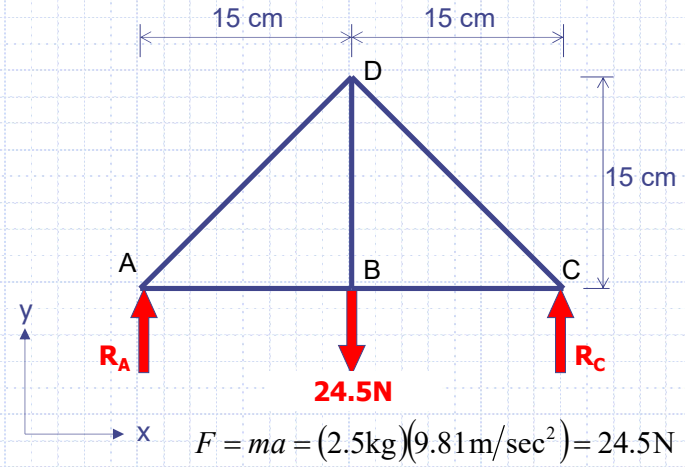
41

Model the Structure



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Draw a Free Body Diagram



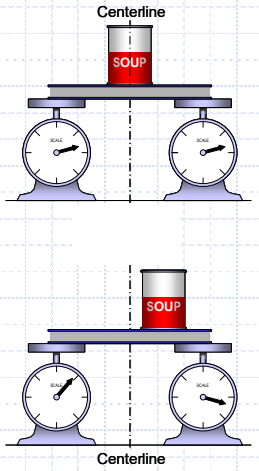
$$F = ma = (2.5\text{kg})(9.81\text{m/sec}^2) = 24.5\text{N}$$

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Calculate Reactions

- Total downward force is 24.5 N.
- Total upward force *must be* 24.5 N.
- Loads, structure, and reactions are all symmetrical.

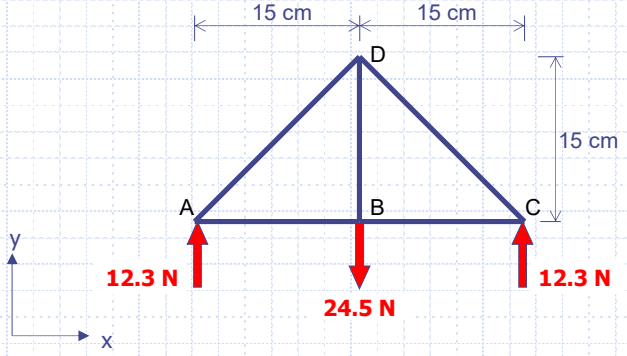
R_A and R_C must be equal.



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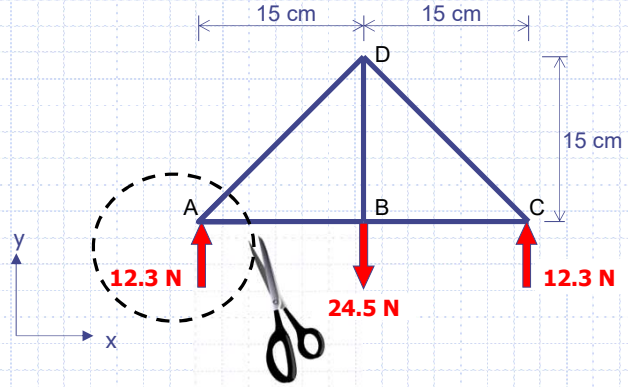
Calculate Reactions

$$R_A = R_C = \frac{24.5}{2} = 12.3\text{N}$$



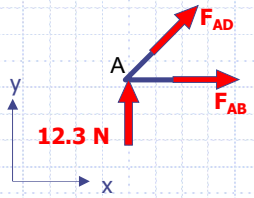
Method of Joints

- Isolate a Joint.



Method of Joints

- Isolate a Joint.
- Draw a free body diagram of the joint.
  - Include any external loads of reactions applied at the joint.
  - Include unknown internal forces at every point where a member was cut.
  - Assume unknown forces in tension.
- Solve the Equations of Equilibrium for the Joint.



EXTERNAL FORCES and INTERNAL FORCES Must be in equilibrium with each other.

Equations of Equilibrium

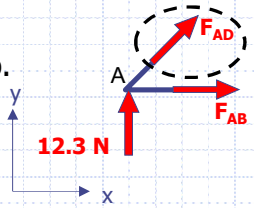
- The sum of all forces acting in the x-direction must equal zero.

$$\sum F_x = 0$$

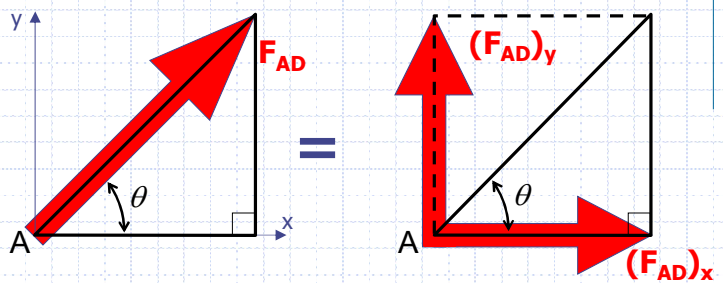
- The sum of all forces acting in the y-direction must equal zero.

$$\sum F_y = 0$$

- For forces that act in a diagonal direction, we must consider both the x-component and the y-component of the force.



Components of Force

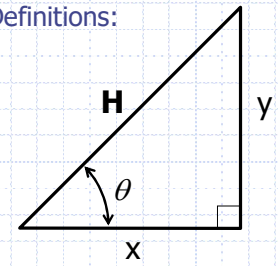


- If magnitude of F_{AD} is represented as the hypotenuse of a right triangle...
- Then the magnitudes of $(F_{AD})_x$ and $(F_{AD})_y$ are represented by the lengths of the sides.

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Trigonometry Review

Definitions:



$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} =$$

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} =$$

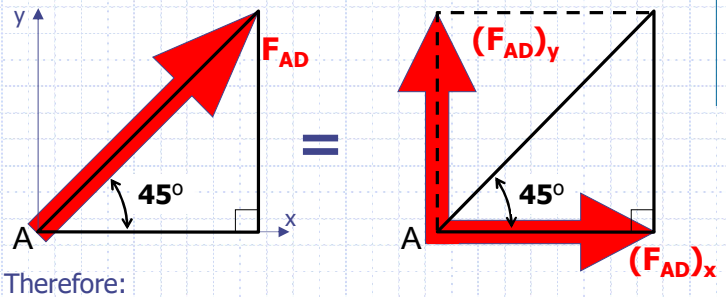
Therefore:

$$x = H\cos\theta$$

$$y = H\sin\theta$$

50

Components of Force



Therefore:

$x = H\cos\theta$ $\quad (F_{AD})_x = F_{AD}\cos 45° = 0.707F_{AD}$

$y = H\sin\theta$ $\quad (F_{AD})_y = F_{AD}\sin 45° = 0.707F_{AD}$

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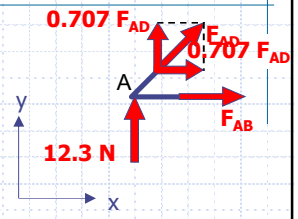
Equations of Equilibrium

$$\sum F_x = 0$$

$$+F_{AB} + 0.707F_{AD} = 0$$

$$F_{AB} = -0.707F_{AD}$$

$$F_{AB} = -0.707(-17.3) = +12.3\text{N}$$



F_{AB}=12.3 N (tension)

$$\sum F_y = 0$$

$$+12.3 + 0.707F_{AD} = 0$$

$$0.707F_{AD} = -12.3$$

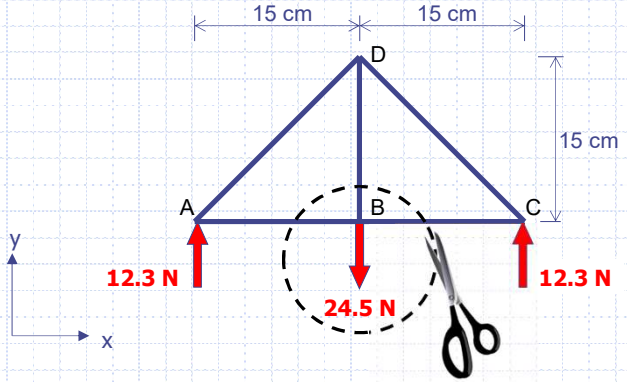
$$F_{AD} = \frac{-12.3}{0.707} = -17.3\text{N}$$

F_{AD}=17.3 N (compression)

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Method of Joints...Again

- Isolate another Joint.



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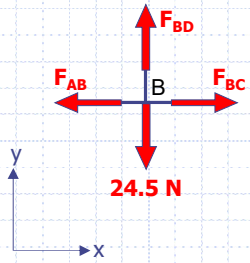
Equations of Equilibrium

$$\sum F_x = 0$$

$$-F_{AB} + F_{BC} = 0$$

$$F_{BC} = F_{AB} = +12.3\text{N}$$

F_{BC}=12.3 N (tension)



$$\sum F_y = 0$$

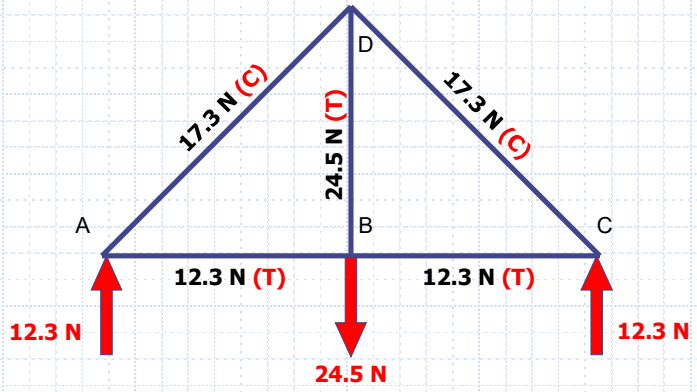
$$-24.5 + F_{BD} = 0$$

$$F_{BD} = +24.5\text{N}$$

F_{BD}=24.5 N (tension)

54

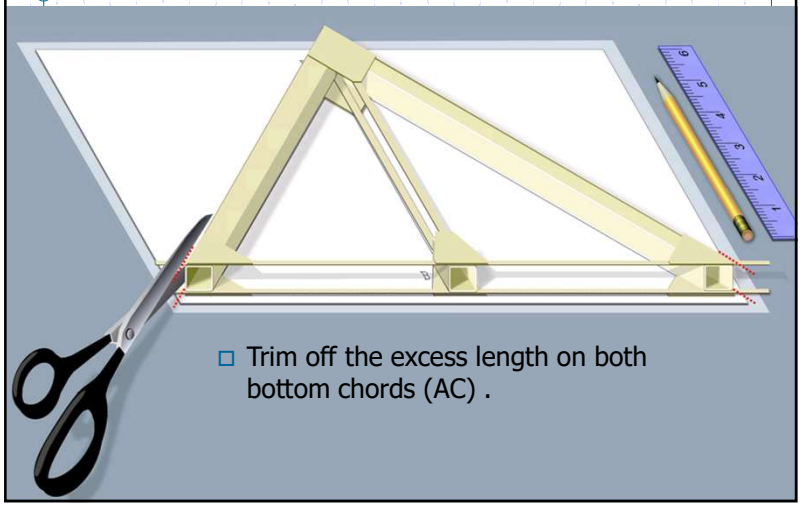
Results of Structural Analysis



Do these results make sense?

55

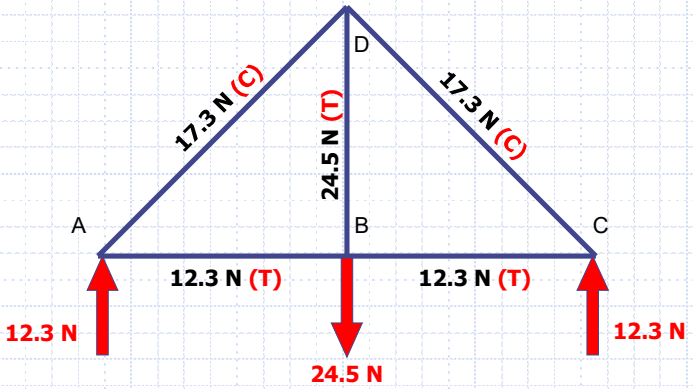
Finish the Truss



- Trim off the excess length on both bottom chords (AC) .

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Results of Structural Analysis



In our model, what kind of members are used for *tension?* for *compression?*

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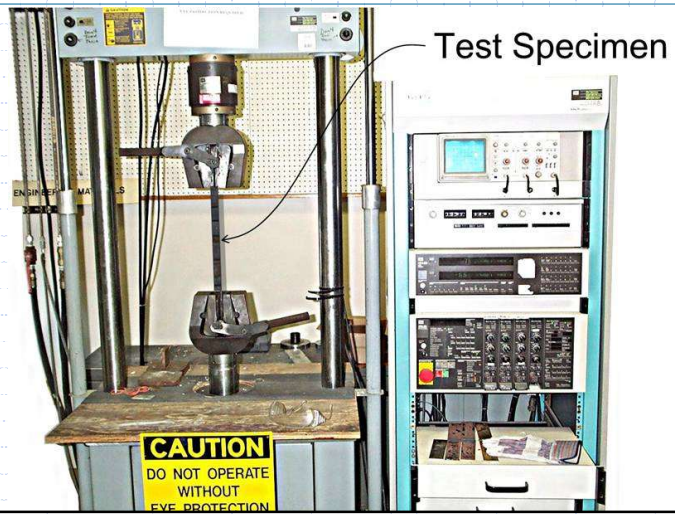
Materials Testing

- Strength – The largest *internal force* a structural member can experience before it *fails*.
- Failure – The condition that occurs when the *internal force* exceeds the *strength* of a member

TENSILE STRENGTH ≠ COMPRESSIVE STRENGTH

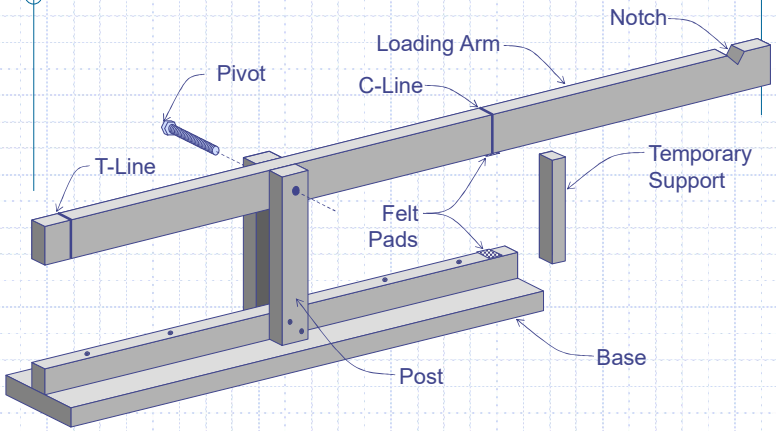
58

A Hydraulic Testing Machine



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Our Low-Budget Testing Machine



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Testing Tensile Strength

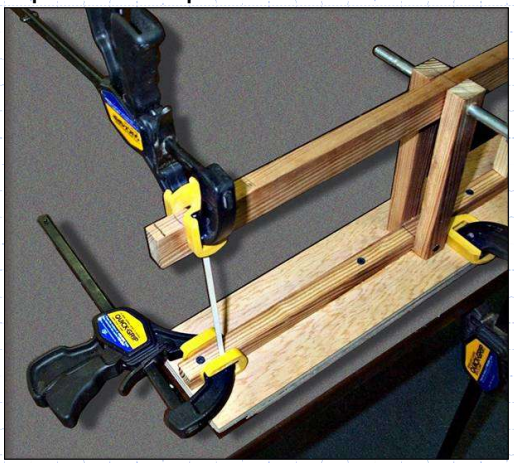
The test setup.



61

Testing Tensile Strength

Clamp the test specimen to the lever arm.



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Testing Tensile Strength

Slowly add sand to the bucket.



63

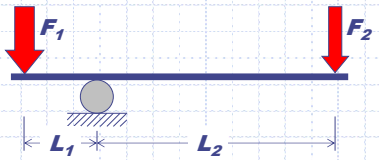
Testing Tensile Strength

When the specimen breaks, weigh the bucket and compute the tensile strength.



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The Principle of the Lever



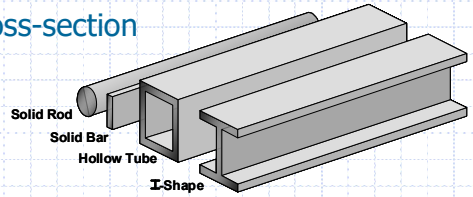
$$F_1 L_1 = F_2 L_2$$

$$F_1 = F_2 \left(\frac{L_2}{L_1} \right)$$

65

Results of Tension Testing

- Tensile strength depends on:
  - Type of material
  - Thickness of cross-section
  - Width of cross-section
- Tensile strength does not depends on:
  - Length of member
  - Shape of cross-section



66

Process the Experimental Results

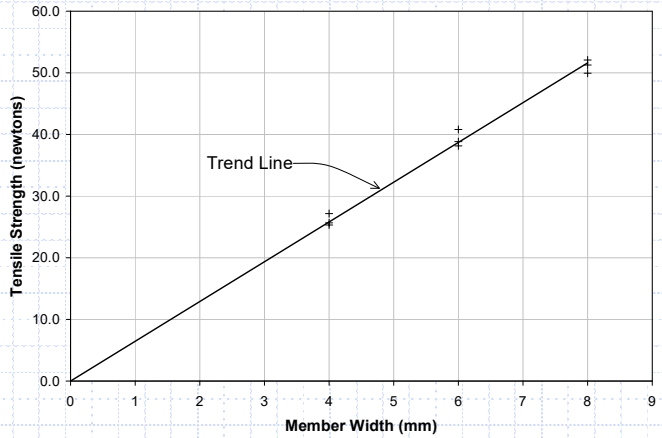
	A	B	C	D	E	F	G
1	L1 =	25		Member Width (mm)	Mass of Bucket & Sand (g)	Weight of Bucket & Sand (N)	Tensile Strength (N)
2	L2 =	69.5					
3							
4				4	942	=E4*9.81/1000	=$B$2*F4/$B$1
5				4	996	=E5*9.81/1000	=$B$2*F5/$B$1
6				4	928	=E6*9.81/1000	=$B$2*F6/$B$1
7				6	1497	=E7*9.81/1000	=$B$2*F7/$B$1
8				6	1424	=E8*9.81/1000	=$B$2*F8/$B$1
9				6	1398	=E9*9.81/1000	=$B$2*F9/$B$1
10				8	1880	=E10*9.81/1000	=$B$2*F10/$B$1
11				8	1909	=E11*9.81/1000	=$B$2*F11/$B$1
12				8	1832	=E12*9.81/1000	=$B$2*F12/$B$1

Convert from grams to newtons

Apply the Principle of the Lever to calculate strength

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Graph the Results



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Testing Compressive Strength

The test setup.



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Testing Compressive Strength

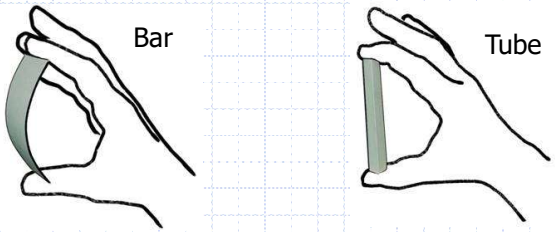
A compression specimen at failure.



70

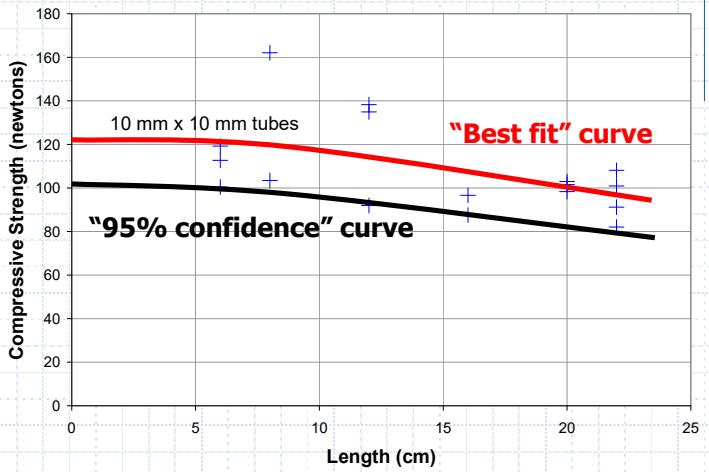
Results of Compression Testing

- Compressive strength depends on:
  - Type of material
  - Length of member
  - Width and thickness of cross-section
  - Shape of cross-section



71

Graph the Results



72

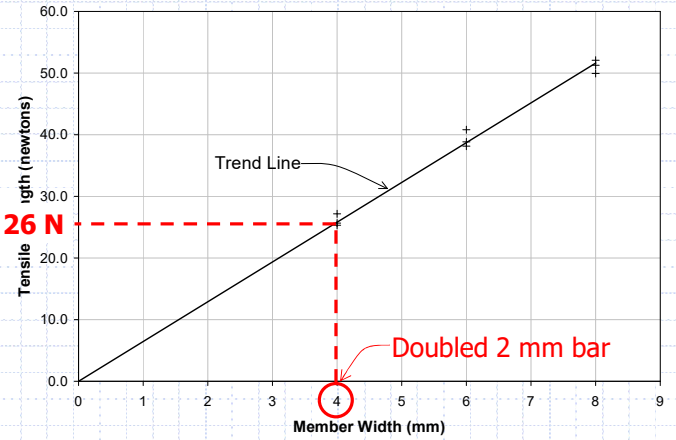
Structural Evaluation

- Is the internal member force less than the strength for each member?
- Calculate the Factor of Safety:

$$\text{Factor of Safety} = \frac{\text{Strength}}{\text{Internal Force}}$$

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Tensile Strength of Member AC



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Factor of Safety for Member AC

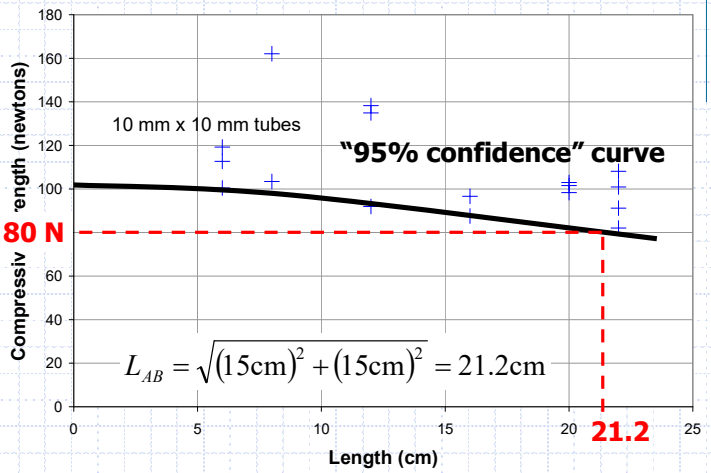
$$\text{Factor of Safety (FS)} = \frac{\text{Strength}}{\text{Internal Force}}$$

$$FS = \frac{26\text{N}}{12.3\text{N}} = 2.1 > 1 \rightarrow \textbf{SAFE!}$$

Structures are normally designed for a FS of at least 1.6.

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Strength of Member AD



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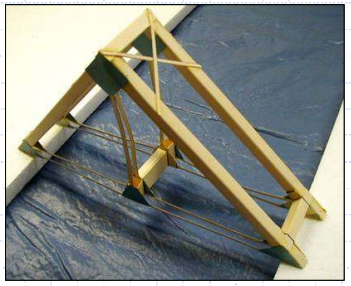
Factor of Safety for AD

$$\text{Factor of Safety (FS)} = \frac{\text{Strength}}{\text{Internal Force}}$$

$$FS = \frac{80N}{17.3N} = 4.6 > 1 \rightarrow \textbf{VERY SAFE!}$$

Are the end posts excessively strong?

Place the Structure into Service



The completed bridge

Load test with 5 kg of sand suspended from midspan



Structural Design

- Design Requirements:
  - Span, loading, factor of safety
- Decide on truss configuration.
- Perform a structural analysis.
  - Reactions
  - Internal member forces
- Select member sizes based on required strength.
- Draw plans.
- Build the bridge.
- Test – *Can the bridge carry the required loading safely?*



Please don't break the bridge!

The West Point Bridge Designer

- Look and feel of a standard CAD package.
- Easy to create a successful design.
- Hard to create a highly competitive design.
- Highly successful:
  - Over 150,000 copies downloaded since 2000.
  - Two major national software awards.
  - Formally endorsed as an educational tool by the American Society of Civil Engineers.
- Runs on Windows 95 (or later) PC.

The West Point Bridge Design Contest

- Started on January 8, 2004.
- Students age 13 through grade 12 are eligible for prizes.
- To enter:
  - Use the **West Point Bridge Designer 2004** to design a bridge.
  - Upload the design to our website for automated judging.
  - Receive instant feedback about contest standing.
- $15,000 scholarships for the winners.
- Participation is **free**!



West Point Bridge Design Contest
bridgecontest.usma.edu

Summary

- File-folder bridges:
  - Accurate representation of real bridges
  - Vehicle for learning engineering concepts.
  - Design based on authentic applications of math, science, and computer technology.
- The West Point Bridge Designer:
  - Experience the engineering design process.
  - Free!
- The West Point Bridge Design Contest:
  - Please help us make it successful!

West Point Bridge Design Contest
bridgecontest.usma.edu